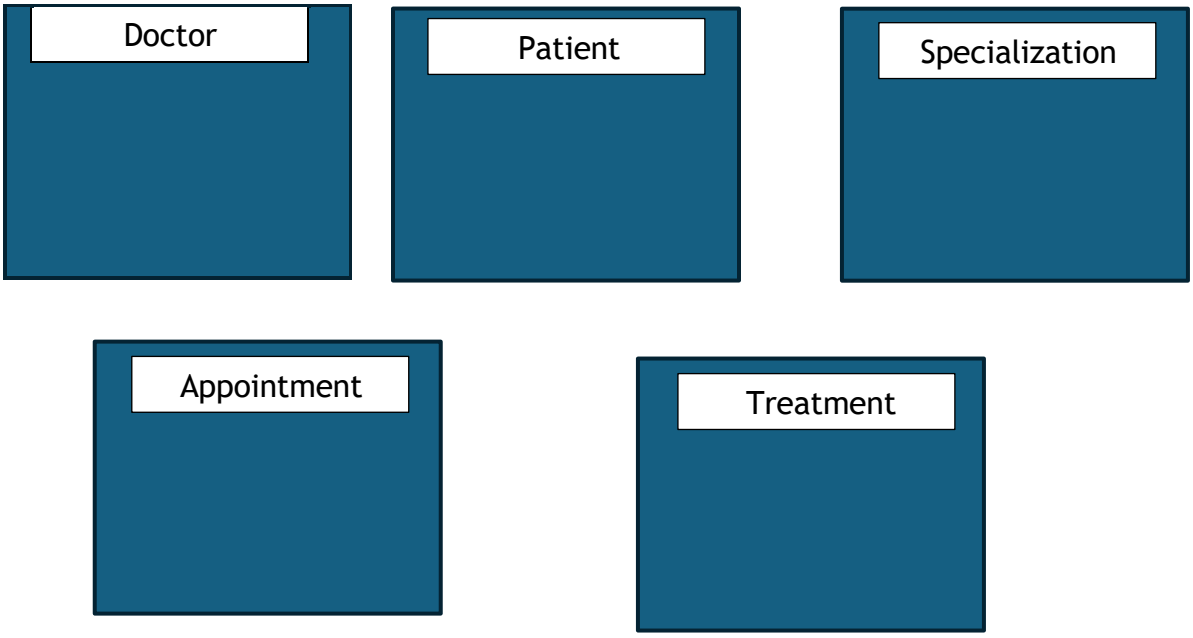
DATA1046B - (GROUP) ASSIGNMENT I

Prepared by: Alphonse Haba

Rasheedat Abdulazeez

Part 1: Entity identification:



Part 3: ER Diagram in MYSQL workbench (Relationship Identification)

- Doctor: One-To-Many: patients
- Patient: one-To-One (mandatory): Doctor

} Doctor: One-To-Many: Patient

- Patient: One-To-Many: appointments
- Appointment: One-To-one (mandatory): Patient

} Patient: One-To-Many: Appointment

- Doctor: One-To-Many: Appointment
- Appointment: One-To-one (mandatory): Doctor

} Doctor: One-To-Many : Appointments

- Doctor: One-To-One: specialization
- Specialization: One-To-Many: Doctors

} Specialization: One-To-Many: Doctor

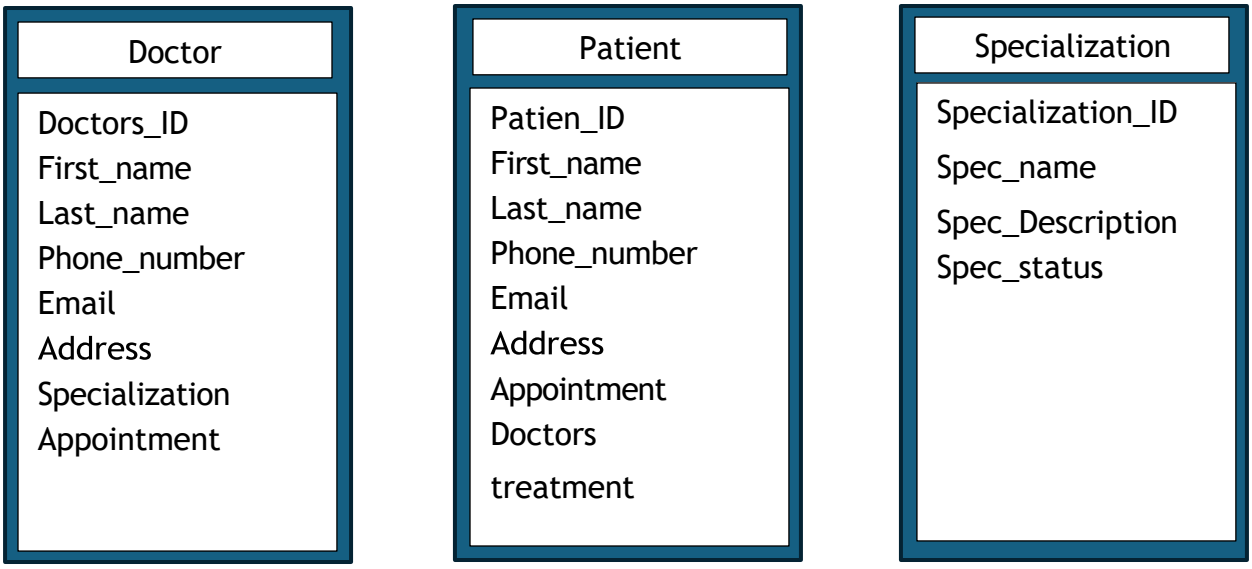
- Patient: One-To-Many: Treatments
- Treatment: One-To-Mant: Patients

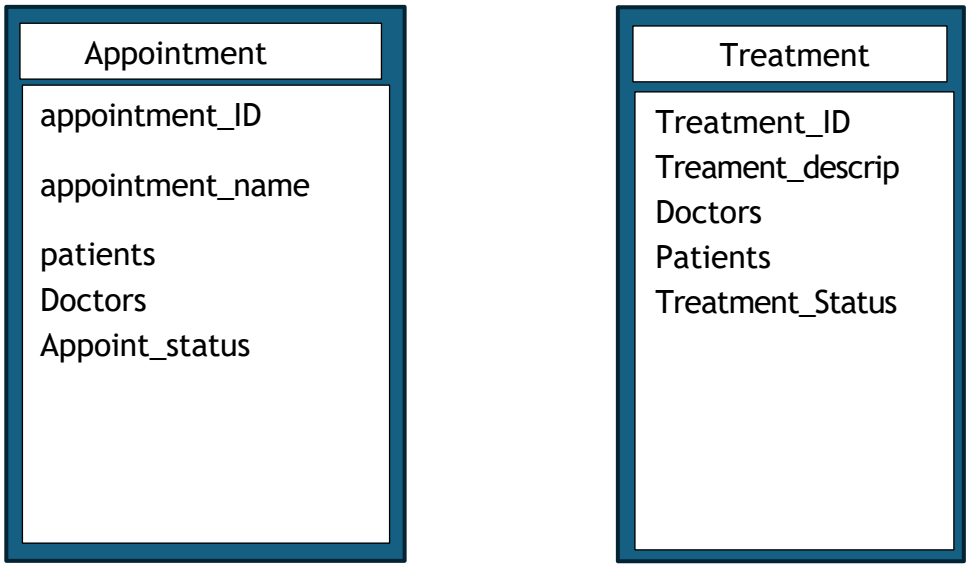
} Patient: One-To-Many: Treatments

- Doctor: One-To-Many: Treatments
- Treatment: One-To-Many: Doctors

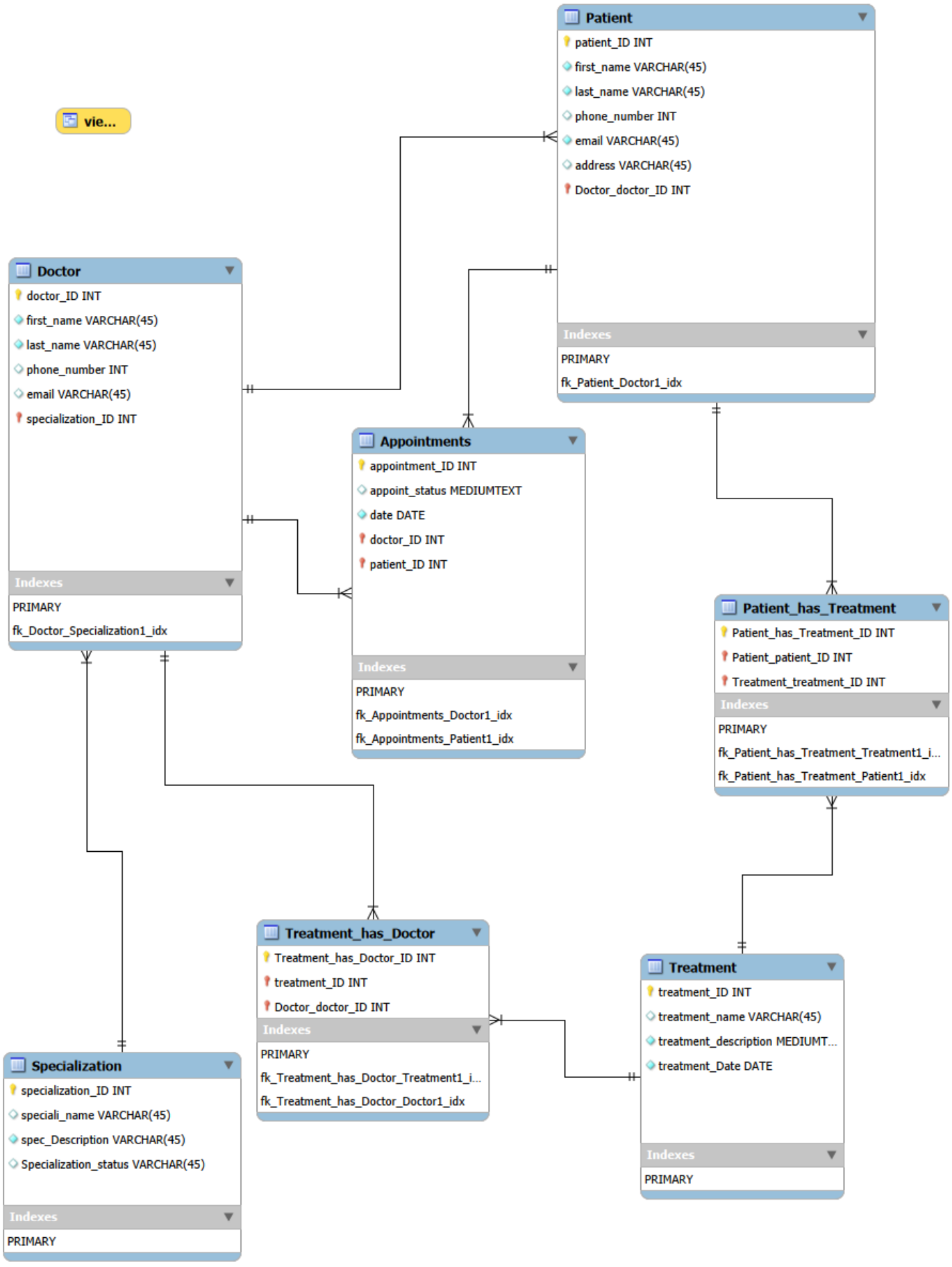
} Doctor: One-To-Many: Treatments

Part 2: non-key attributes identifications





5) Crow foot notation in ER diagram in MySQL workbench


vie...
Patient
patient_ID INT
first_name VARCHAR(45)
last_name VARCHAR(45)
phone_number INT
email VARCHAR(45)
address VARCHAR(45)
Doctor_doctor_ID INT
Indexes
PRIMARY
fk_Patient_Doctor1_idx
Doctor
doctor_ID INT
first_name VARCHAR(45)
last_name VARCHAR(45)
phone_number INT
email VARCHAR(45)
specialization_ID INT
Indexes
PRIMARY
fk_Doctor_Specialization1_idx
Appointments
appointment_ID INT
appoint_status MEDIUMTEXT
date DATE
doctor_ID INT
patient_ID INT
Indexes
PRIMARY
fk_Appointments_Doctor1_idx
fk_Appointments_Patient1_idx
Patient_has_Treatment
Patient_has_Treatment_ID INT
Patient_patient_ID INT
Treatment_treatment_ID INT
Indexes
PRIMARY
fk_Patient_has_Treatment_Treatment1_i...
fk_Patient_has_Treatment_Patient1_idx
Treatment_has_Doctor
Treatment_has_Doctor_ID INT
treatment_ID INT
Doctor_doctor_ID INT
Indexes
PRIMARY
fk_Treatment_has_Doctor_Treatment1_i...
fk_Treatment_has_Doctor_Doctor1_idx
Treatment
treatment_ID INT
treatment_name VARCHAR(45)
treatment_description MEDIUMT...
treatment_Date DATE
Indexes
PRIMARY
Specialization
specialization_ID INT
speciali_name VARCHAR(45)
spec_Description VARCHAR(45)
Specialization_status VARCHAR(45)
Indexes
PRIMARY

```
CREATE DATABASE IF NOT EXISTS myhospitaldb;
USE myhospitaldb;

-- 1. Specialization table with primary key
CREATE TABLE IF NOT EXISTS Specialization (
   specialization_ID INT NOT NULL PRIMARY KEY,
   specialization_name VARCHAR(45) NOT NULL,
   Spec_Description VARCHAR(45) NOT NULL,
   specia_status VARCHAR(45) NOT NULL
);

-- 2. Doctor table : one-to-many relationship with specialization table.
-- primary and foreign key to represent entity in other table.
CREATE TABLE IF NOT EXISTS Doctor (
   doctor_ID INT NOT NULL PRIMARY KEY,
   first_name VARCHAR(45) NOT NULL,
   last_name VARCHAR(45) NOT NULL,
   phone_number VARCHAR(20) NOT NULL,
   email VARCHAR(255) NOT NULL,
   address VARCHAR(45),
   specialization_ID INT NOT NULL,

   CONSTRAINT fk_doctor_specialization
      FOREIGN KEY (specialization_ID)
      REFERENCES Specialization(specialization_ID)
);

-- 3. Patient table: One-To-Many relationship with primary and foreign key from doctor table.
CREATE TABLE IF NOT EXISTS Patient (
   patient_ID INT NOT NULL PRIMARY KEY,
   first_name VARCHAR(45) NOT NULL,
   last_name VARCHAR(45) NOT NULL,
   email VARCHAR(255) NOT NULL,
   phone_number VARCHAR(20) NOT NULL,
   address VARCHAR(45),
   doctor_ID INT NOT NULL,

   CONSTRAINT fk_patient_doctor
      FOREIGN KEY (doctor_ID)
      REFERENCES Doctor(doctor_ID)
);

-- 4. Appointment table : with one-to-many relationship . foreign keys to represent
-- others attribute from doctor and patient
CREATE TABLE IF NOT EXISTS Appointment (
   appointment_ID INT NOT NULL PRIMARY KEY,
   appoint_status MEDIUMTEXT,
   appoint_date DATE NOT NULL,
   patient_ID INT NOT NULL,
   doctor_ID INT NOT NULL,

   CONSTRAINT fk_appointment_patient
```

```
      FOREIGN KEY (patient_ID)
      REFERENCES Patient(patient_ID),

   CONSTRAINT fk_appointment_doctor
      FOREIGN KEY (doctor_ID)
      REFERENCES Doctor(doctor_ID)
);

-- 5. Treatment table with primary key
CREATE TABLE IF NOT EXISTS Treatment (
   treatment_ID INT NOT NULL PRIMARY KEY,
   treatment_name VARCHAR(45) NOT NULL,
   treatment_descrip MEDIUMTEXT NOT NULL,
   treatment_date DATE NOT NULL
) ;

-- 6. Relationship Many-To-Many with bridge table : Doctor and Treatment
CREATE TABLE IF NOT EXISTS doctor_bridge_treatment (
   doctor_treatment INT NOT NULL PRIMARY KEY,
   doctor_ID INT NOT NULL,
   treatment_ID INT NOT NULL,

   CONSTRAINT uq_doctor_treatment
      UNIQUE (doctor_ID, treatment_ID),

   CONSTRAINT fk_doctor_treatment_bridge
      FOREIGN KEY (doctor_ID)
      REFERENCES Doctor(doctor_ID),

   CONSTRAINT fk_treatment_doctor_bridge
      FOREIGN KEY (treatment_ID)
      REFERENCES Treatment(treatment_ID)
);

-- 6. Relationship Many-To-Many with bridge table : Patient and Treatment
CREATE TABLE IF NOT EXISTS patient_treatment_bridge (
   patient_treatment INT NOT NULL PRIMARY KEY,
   patient_ID INT NOT NULL,
   treatment_ID INT NOT NULL,

   CONSTRAINT uq_patient_treatment
      UNIQUE (patient_ID, treatment_ID),

   CONSTRAINT fk_patient_treatment_bridge
      FOREIGN KEY (patient_ID)
      REFERENCES Patient(patient_ID),

   CONSTRAINT fk_treatment_patient_bridge
      FOREIGN KEY (treatment_ID)
      REFERENCES Treatment(treatment_ID)
);
```